Rank of matrix:

Rank = no of non-zero row in upper triangular matrix

Linear transformation:

we have a rule T that takes a vector from 1 vector space (V) and gives us another vector space (W).

$T(kv) = kT(v)$

↓ multiply first transform later

↓ transform first multiply later

$T(5v) = 5T(v)$

$v = (2,3)$ $x = 5$

$T(u+v) = T(u) + T(v)$

meeting these two conditions only then T is linear

Note: If we apply a transformation and it takes a vector and returns in the same space

$V = W$

e.g Rotation in $\mathbb{R}^2 \to \mathbb{R}^2$ ∴ in this case we call (T) a linear operator.

Notation:

$$T: V \to W$$

Q $T(x,y) = (2x-y,\ 3y)$ $\quad T: \mathbb{R}^2 \to \mathbb{R}^2$

(a) Show T is a linear transformation

$T(ku) = kT(u)$

$u = (x_1, y_1)$

$T(k(x_1, y_1)) = kT(x_1, y_1)$

$T(kx_1, ky_1) = k(x_1, y_1)$

$(kx_1, ky_1) = (kx_1, ky_1)$

$k(2x-y, 3y) = k(2x-y, 3y)$ ✓

$T(u+v) = T(u) + T(v)$

let $u = x_1, y_1 \quad v = x_2, y_2$

$T(x_1+x_2, y_1+y_2) = T(x_1, y_1) + T(x_2, y_2)$

$T(2x-y+2x-y,\ 3y+3y) = T(2x-y, 3y) + T(2x-y, 3y)$

$T(4x-2y, 6y) = T(2x-y, 3y) + T(2x-y, 3y)$

$(4x-2y, 6y) = (2x-y, 3y) + (2x-y, 3y)$

$(4x-2y, 6y) = (4x-2y, 6y)$ ✓

(b) find $T(0,0)$, $T(u-v)$ where $u = (3,1)$ $v = (1,5)$ and $T(-u)$

$T(0,0) \to (0,0)$

$u = (3,2)$, $v = (1,5)$ $\quad (3,2) - (1,5)$

$\rightarrow (2,-3)$

$T(2,-3) = (2(2) - (-3),\ 3(-3))$

$= (7,-9)$ ✓

$T(-3,-2) = (2(-3) - (-1),\ 3(-2))$

$= (-6+2,\ -6)$

$= (-4,-6)$

Identity operator:

$I : V \rightarrow V$

↓

I is linear

This transformation does nothing to the vector it just returns the same vector.

1) $I(u+v) = I(u) + I(v)$

2) $I(ku) = kI(u)$

(Dilation and contraction operators)

$T(x) = c \cdot x$ → vector

↓

any no

↓

T just multiplies vector by a c

$0 < c < 1$ → contraction (vector gets smaller)

$c > 1$ dilation (vectors get big)

and $T(x) = c \cdot u$ is again Linear.

A linear transformation for polynomials $T: P_n \to P_{n+1}$ (→ degree)

$$T(p(x)) = x \cdot p(x)$$

e.g $p(x) = 1 + 2x$

$$T(p(x)) = x(1 + 2x) = x + 2x^2$$

A linear transformation for inner product spaces

$T: V \to \mathbb{R}$ $\quad T(x) = \langle x, v_0 \rangle$

$$T(ku) = \langle ku, v_0 \rangle$$
$$T(u+v) = \langle u+v, v_0 \rangle$$

$v_0 \to$ fixed vector

e.g 10?

Q linear transformation using inner product.

$v_0 = (2,3)$ $\quad T(x) = \langle x, v_0 \rangle$ Transformation $T(x) = \langle x, v_0 \rangle$

Chosing an vector : $x = (1,4)$

$$T(x) = \langle (1,4), (2,3) \rangle$$
$$= 2 + 12 = \boxed{14}$$

1) scaling $k=3$ $\quad T(ku) = \langle (3,12), (2,3) \rangle$

$$= 6 + 36 \to \underline{42}$$

compare with $kT(u) = 14 \times 3 = 42$

2) Adding two vectors. Let $q = (1,2)$

$$T(u+q) = T(u) + T(q)$$

Kernel of Linear transformation

aka ↓ $T: V \to W$

nullspace

The kernel is the set of all vectors that disappear (become 0) when we apply T.

e.g

$$T(x,y) = (x-y, 0)$$

find kernel

$x - y = 0 \quad 0 = 0$

$x - y$

So All vectors of form (x,y)

$= (t,t)$ $t \in$ Real no.

$(3,3)$

$T(3,3) = (3-3, 0) = (0,0)$

find kernel for $T(x,y,z) = (x,y)$

$T(x,y,z) = (0,0,0)$

$(x,y) = (0,0)$

$x = 0, \quad y = 0 \quad z = t$ treating z as free var

kernel is $(0,0,t)$

e.g $(0,0,-7)$

$T(0,0,-7) = (0,0)$

nullity = n(T) is the no of elements in the kernel Set. Non-zero elements. It is also the dimension of kernel of T. | no of free var ↓ variables that don't have fixed value

e.g $T(x,y,z) = (x+y, y+z)$

find kernel

$T(x,y,z) = (0,0,0)$

$(x+y, y+z) = (0,0)$

$x+y=0 \quad y+z=0$ Suppose $z=t$ treating it as a free variable

$x=-y \quad y=-z$

$y=-t \quad z=t$

$x=-(-z)$

$x=z \rightarrow t$

kernel $\rightarrow (t,-t,t) \rightarrow t(1,-1,1)$

nullity = 1 as z is the only free variable

Range: $T: V \rightarrow W$

↳ is all vectors in W that T can produce

e.g $T(x,y) = (x+y, x+y)$

find all outputs $(x+y, x+y) = (a,a)$ ← can by any real no.

$R(T) = (a,a)$

kernel = inputs that give 0

Range = outputs of T.

Dimension of R(n) / Range is called Rank.

Q consider the following transformation:

$$T(x, y) = (x+y, x-y, y)$$

i) nullspace $T(x, y) = (0, 0)$

$x+y=0$ $\quad x-y=0$ $\quad y=0$

$x=0$ $\quad x=0$ $\quad y=0$

nullspace / kernel $= \{0, 0\}$

nullity $= 0$. no free var.

ii) Range $T(x, y) = (x+y, x-y, y)$

write T as a matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 1 \end{pmatrix}$$

finding its rank aka dimension of range is the no of non-zero rows.

$$T(x, y) = (x+y, x-y, y)$$

$$= (a, b, c)$$

$a = x+y$ $\quad b = x-y$ $\quad c = y$

starting with $y = c$

finding x

$b = x-y$

$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ Rank = 2

$b = x - c$

$x = b + c$ (2)

$y = c$ (1)

$a = b + c + c$ (3)

$a = b + 2c$ → final eq.

Just plug in every eq. into each other

Rank + nullity = dim of V

$T(x, y) = (x + y, x - y, y)$

$T : V \to W$

V: $\mathbb{R}^2$

W: $\mathbb{R}^3$

(compositions and inverse Transformations)

$T: V \to W$ is one-one if for diff inputs we get diff outputs.

$T(x) = 2x$ $\quad T(1) = 2$ $\quad T(2) = 4$

↓

one to one

2) onto $T: V \to W$??

if kernel is 0 that means the transformation is zero.

kernel = $\{0\}$ → one to one

→ Rotation T in $\mathbb{R}^2$ is both one-one and onto.

→ Differentiation is not one to one

→ Projections in $\mathbb{R}^2$ are not one to one | onto